

1. A 28-year-old mountain climber reaches a high-altitude base camp where the partial pressure of oxygen is significantly lower than at sea level. Which of the following statements regarding his physiological adaptation is CORRECT?

A. His red blood cells will decrease the synthesis of 2,3- β PG to hold onto oxygen more tightly.

B. There will be an initial leftward shift of the oxygen dissociation curve due to hyperventilation, followed by a rightward shift caused by increased 2,3- β PG.

C. The concentration of hydrogen ions in his red blood cells will increase, directly stimulating more 2,3- β PG production.

D. High altitudes cause 2,3- β PG to bind more effectively to the "R" (relaxed) state of hemoglobin.

Answer: B

Explanation: At high altitudes, low oxygen levels induce hyperventilation, which lowers $p\text{CO}_2$ and hydrogen ions, causing an initial leftward shift of the oxygen dissociation curve. To compensate and ensure oxygen is delivered to tissues, the red blood cells increase the production of 2,3- β PG. This 2,3- β PG binds to the central pocket of deoxyhemoglobin, stabilizing the T-state and shifting the curve back to the right, facilitating oxygen unloading. Statement C is False because the relationship between H^+ and 2,3- β PG is inversely proportionate.

2. A patient is brought to the emergency room with suspected carbon monoxide (CO) poisoning after being found in a garage with a running car. Which of the following regarding carboxyhemoglobin (COHb) is NOT CORRECT?

A. CO has approximately 200 times higher affinity for hemoglobin than oxygen.

B. The binding of one CO molecule to a hemoglobin tetramer increases the oxygen affinity of the remaining heme sites.

C. CO toxicity causes a significant rightward shift in the oxygen dissociation curve, making it easier for tissues to receive oxygen.

D. Clinical manifestations are typically observed when COHb levels exceed 20%.

Answer: C

Explanation: This statement is Not Correct. While CO binds to the same sites as O₂ with much higher affinity, its binding actually causes a leftward shift in the dissociation curve for the remaining oxygen molecules. This means that while hemoglobin might be carrying some oxygen, it refuses to "unload" it to the tissues, leading to cellular suffocation. Statements A, B, and D are True based on the clinical properties of COHb.

3. A physician is monitoring a patient with Type 2 Diabetes Mellitus to assess their long-term blood glucose control over the past 2–3 months. Which hemoglobin derivative provides the most accurate prognostic information?

A. Methemoglobin

B. Carboxyhemoglobin

C. Carbaminohemoglobin

D. Glycosylated hemoglobin (HbA1c)

Answer: D

Explanation: Correct. HbA1c is produced when glucose reacts with the amino groups (specifically the N-terminal valines of the β -chains) of hemoglobin to form a ketoamine. Because red blood cells live for about 120 days, the level of HbA1c reflects the average blood sugar levels over that period, making it a vital tool for the prognosis and prediction of diabetic complications.

4. A pregnant woman at 30 weeks gestation is at risk of preterm labor. The medical team performs an amniocentesis to check the Lecithin–Sphingomyelin (L/S) ratio. Which of the following findings would indicate a high risk for Infant Respiratory Distress Syndrome (IRDS)?

A. An L/S ratio of 2.5.

- B. An L/S ratio of 1.2.
- C. Increased presence of SP-A and SP-D in the amniotic fluid.
- D. Rapid accumulation of DPPC in Alveolar Type II cells.

Answer: B

Explanation: True. For predicting pulmonary lung maturity, the critical L/S ratio is 2.0 or greater. A ratio below 2.0 (like 1.2) is a significant risk factor for IRDS, as it indicates insufficient surfactant (specifically lecithin/DPPC) to prevent alveolar collapse. If the test is positive for risk, glucocorticoids may be given to the mother to accelerate lung maturation.

5. A child is diagnosed with Sickle Cell Anemia (HbS) after presenting with a painful vaso-occlusive crisis. Which of the following describes the molecular basis of this disease?

- A. A mutation where lysine replaces glutamate at the 6th position of the β -chain.
- B. A complete lack of α -chain synthesis leading to hemolysis.
- C. A mutation where valine replaces glutamic acid at the 6th position of the β -chain.
- D. The oxidation of ferrous iron (Fe^{2+}) to ferric iron (Fe^{3+}) in the heme group.

Answer: C

Explanation: Correct. Sickle-cell anemia (HbS) is the most common abnormal hemoglobin where glutamic acid is replaced by valine at the 6th position of the β -chain. This single amino acid change causes the hemoglobin to polymerize and "sickle" under low oxygen conditions, leading to the clinical symptoms of the disease. Statement A describes Hemoglobin C disease, not HbS.

6. All of the following are components of Protoporphyrin IX EXCEPT:

- A. 4 Pyrrole rings
- B. 4 Methene bridges
- C. 1 Magnesium ion (Mg^{2+})
- D. 2 Propionate side chains

Answer: C

Explanation: Correct choice for "Except." Protoporphyrin IX contains 4 pyrrole rings, 4 methene bridges, 4 methyl groups, 2 vinyl groups, and 2 propionate side chains. Magnesium is found in chlorophyll, whereas the heme in hemoglobin requires Ferrous iron (Fe^{2+}).

7. Regarding the location of myoglobin and hemoglobin, which of the following is TRUE?

- A. Hemoglobin is found primarily in the striated muscle cytoplasm.
- B. Myoglobin is restricted exclusively to the erythrocytes.
- C. Myoglobin is found primarily in skeletal and striated muscle.
- D. Both proteins are found in equal concentrations in the plasma.

Answer: C

Explanation: True. Myoglobin is a monomeric protein found in skeletal and striated muscle, where it serves as an oxygen store. Hemoglobin is restricted to erythrocytes and is responsible for transporting oxygen from the lungs to other tissues.

8. Which level of protein organization refers to the assembly of two or more separate polypeptide chains held together by non-covalent interactions?

- A. Primary structure

B. Secondary structure

C. Tertiary structure

D. Quaternary structure

Answer: D

Explanation: Correct. Quaternary structure is the assembly of multiple subunits (polypeptide chains). Hemoglobin has a quaternary structure because it is a tetramer (two α and two β chains), whereas Myoglobin is a monomer and lacks quaternary structure.

9. The Bohr Effect describes how specific factors decrease hemoglobin's affinity for oxygen. Which of the following shifts the oxygen dissociation curve to the right?

A. Increased pH (alkalinity)

B. Decreased pCO_2

C. Increased concentration of H^+ (lower pH)

D. Decreased temperature

Answer: C

Explanation: True. The Bohr Effect states that increased H^+ (lower pH) and increased pCO_2 decrease oxygen affinity, shifting the curve to the right to facilitate oxygen delivery to metabolically active tissues.

10. Which state of hemoglobin is characterized by low oxygen affinity and is stabilized by the binding of 2,3-BPG?

A. R (Relaxed) state

B. T (Taut/Tense) state

C. Hyperbolic state

D. Secondary state

Answer: B

Explanation: Correct. The T-state (deoxyhemoglobin) has a low affinity for oxygen. 2,3-BPG specifically binds to and stabilizes the T-state by forming ionic salt bridges between the β -chains, which encourages the release of oxygen.

11. Which of the following statements regarding the proteins in pulmonary surfactant is NOT CORRECT?

A. Surfactant-specific proteins SP-A and SP-D are collectins that promote phagocytosis of bacteria and viruses by macrophages.

B. SP-B and SP-C are hydrophobic membrane proteins that increase the rate at which surfactant spreads over the alveolar surface.

C. Surfactant proteins SP-A, SP-B, SP-C, and SP-D make up approximately 90% of the total surfactant composition.

D. SP-B and SP-C are hydrophobic proteins required for the proper biophysical function of the lung.

Answer C

12. All of the following statements regarding stored transfusion blood are correct EXCEPT:

A. Stored blood loses vital 2,3-BPG over time.

B. Reconstitution with inosine prevents the decrease of 2,3-BPG.

C. Stored blood undergoes a rapid increase in 2,3-BPG levels during storage.

D. Inosine added to stored blood contains hypoxanthine-ribose.

Answer: C

13. Which of the following is TRUE regarding the physiological impact of a minor pH drop from 7.4 to 7.2?

A. It decreases oxygen unloading efficiency by 11%.

B. It has no effect on oxygen unloading.

C. It increases oxygen unloading efficiency by 11%.

D. It shifts the oxygen dissociation curve to the left

Answer: C